

The field, the museum and the lecture hall: the spaces of natural history in Victorian Cornwall

Simon Naylor

This paper contributes to our understandings of the geographies of science through an analysis of nineteenth-century natural history and, in particular, of the provincial natural history society. Focusing on nineteenth-century Cornwall and one of the main natural history societies operating in the county at that time – the Penzance Natural History and Antiquarian Society – it is argued that a set of key spaces were integral to the operations and outputs of such societies. The paper details the significance of the Penzance Society's museum, field sites and lecture hall as sites for communal work of local natural historians. They were also important, it is argued, in their construction of West Cornwall as a site of national natural scientific importance. Lastly, these spaces defined an agenda for regional scientific study. In particular, they promoted a taxonomic method that would transform local people into rigorous scientists and the local region into a 'book of nature'.

key words Cornwall natural history region geographies of science taxonomy

School of Geographical Sciences, University of Bristol, Bristol BS8 1SS
email: simon.naylor@bristol.ac.uk

revised manuscript received 23 May 2002

Introduction: science's geographies

In the history of geography and in the history of science, it has become increasingly common to note that science has a geography. Leading commentators across a range of disciplines have, in recent years, provided us with leads into some of the ways in which we might grasp science's spaces and spatialities (see, for instance, [Ophir and Shapin 1991](#); [Livingstone 1995](#); [Crang 1998](#); [Shapin 1998](#); [Smith and Agar 1998](#)). Such interventions have had a number of notable effects, the most general being the negation of the thesis that scientific thought and practice are somehow disembodied and universal. In the words of Shapin:

Science was not one thing – conceptually and methodologically unified . . . it was a variety of practices whose conceptual identities were the outcomes of *local patterns* of training and socialization. ([Shapin 1998](#), 6, my emphasis)

Scientific knowledge, it is now assumed, is and always has been locally produced; constructed, contested, negotiated and consumed by a group of interested parties across a complex array of sites, from the laboratory to the drawing room, from the jungle to the zoo. Indeed, it seems now to be generally agreed, amongst science studies students at least, that *where* science is done affects the reliability, veracity and very shape of the knowledge produced. In this vein, Withers has asserted that that we should not

just claim that . . . science has a geography, but that we [should] show how and where and do so by grounding empirical work in appropriate historical and theoretical context. ([Withers 1998](#), 68)

This leads us on to the second, and more specific, aspect of this spatial turn in science studies: the investigation of the places and times in which science has been made, and where it has in turn

re-made the world. Thrift, Driver and Livingstone have highlighted a bewilderingly diverse array of sites of scientific knowledge production (laboratories, lecture halls, observatories, museums and map rooms, for instance), a range of sites where scientific knowledge might be constructed at a distance (scientific societies and associations, corresponding societies, lecture tours, exhibitions, scientific publications), a cornucopia of technologies of knowledge dissemination and presentation, and a set of spaces where these knowledges can be legitimately collected (Thrift *et al.* 1995, 13). Researchers across a panoply of disciplines have begun to research the historical geographies of these spaces. Geographers, historians, anthropologists, sociologists and others have provided detailed expositions of laboratories, physic gardens, field sites, museums, scientific societies and institutions, the home, the body and even the pub, as sites of scientific practice (for instance, Stafford 1995; Bennett and Mandelbrote 1998; Driver 2000; Kohlstedt 1988; Pyenson and Sheets-Pyenson 1998; Shapin 1988; Lawrence and Shapin 1998; Secord 1994a, respectively; for a fuller bibliography of studies into the spaces of science, see Livingstone 2000 2002).

This paper contributes to our understandings of the geographies of science through an analysis of nineteenth-century natural history and, in particular, of the provincial natural history society. There has been a consistent, if not voluminous, dissemination of research on provincial natural history societies in Britain over the last 30 years. This body of work has demonstrated the significance of these societies in, for example, the establishment of professional scientific identities and audiences (Alberti 2001), the promotion of public involvement in and understandings of science (Shapin 1972), the development of the collections of metropolitan gentlemen scientists (Secord 1994b), the creation of new social spaces and relations (Allen 2000b) and in the production of local civic and industrial cultures (Morrell 1983). The aim of this paper is to explore the *geographies* of these societies – their lecture-halls, field sites and museums – and their significance in the production of scientific and civic knowledges of the region. This argument cleaves strongly to Withers' claim, in his own treatment of local natural history societies in nineteenth-century Scotland, that natural history societies

should be understood as local *sites* for making natural and national knowledge, that the members valued *local*

knowledge for its own sake *and* as part of scientific understanding more generally, and that their members' fieldwork and publications should be seen as the means by which given *localities* were examined. (Withers 2001, 187, original emphasis)

In my treatment of natural history in nineteenth-century Cornwall and, in particular, of the main natural history society operating in the county at that time – the Penzance Natural History and Antiquarian Society (PNHAS) – I argue that such societies made heavy use of a range of spaces through which they constructed strong notions of a local natural region and ways of approaching its study, as well as of an understanding of their work's significance within a broader national scientific endeavour. This sort of study is crucial if we are to improve our understanding of the historical geographies of Victorian science, whether those geographies are at the scale of the lecture hall or the region. This localist approach to the writing of the historical geographies of science is also important, in that I believe the only way to understand how science has operated is to follow it through the spaces and times in which it has been performed; to trace the particular and often mundane ways in which it, as a practical and intellectual exercise, has actually been done.

A historical geography of natural history in nineteenth-century Britain

David Allen's *The naturalist in Britain* (1994 [1974]) is probably the most comprehensive and sustained exposition of the social history of British natural history to have been written in recent years. In his canonical text, Allen argues that natural history as a social activity began some time during the course of the seventeenth century and he awards particular significance to the Society of Apothecaries, one of the oldest-established livery companies of the City of London (see also Allen 2000a). This Society required of its apprentices a thorough understanding of drug plants and this was achieved through the provision of physic gardens and, importantly for this history, the organization of fieldcourses into the countryside around London.

Although more sedentary in nature, the founding of the Royal Society of London in 1660 marked another significant step in the production of a scientific community in England, while its unofficial spinoff, the Temple Coffee House Botanic Club (founded in 1698), was probably the earliest

natural history society in the world (Allen 1994). A second wave of enthusiasm for natural history emerged in England and Scotland in the eighteenth century, with a spate of informal clubs being formed in the latter half of the century and culminating in the formation of James Edward Smith's Linnean Society of London in 1788 (Allen 1994; Jenkins 1978). The first decades of the nineteenth century saw the establishment of a series of other influential bodies in England, such as the Geological Society of London in 1807, the Zoological Society of London in 1826 and the Botanical Society of London in 1836. Indeed, it is common parlance for historians of science to refer to this century as the 'Age of Societies' (cf. Turk 1959).

These formal societies, in conjunction with a reinvigoration of natural history in British universities and the effects of the 1815 Apothecaries Act,¹ promoted a more general enthusiasm for natural history among the populace. For instance, it was during the last years of the eighteenth and the early years of the nineteenth century that we see the emergence of a range of comprehensive guidebooks written for amateur naturalists (Larsen 1996; on early modern collecting see Swann 2001). As well as these more generic texts, Withers (2001) has also identified the increased significance, from the early nineteenth century, of natural history volumes aimed at readers and practitioners in particular regions.

During this period, a range of scientific societies beyond the metropolises of London and Edinburgh began to emerge, spreading through the industrial regions of the north of England and southern Scotland. As well as a corpus of artisan natural history societies that clustered around the industrial towns of Lancashire, Yorkshire and the West Midlands, other societies were established in university towns, such as the Ashmolean Natural History Society in Oxford, formed in 1828 (see Allen 1994; Secord 1994a; Alberti 2001). This swath of clubs in the first three decades of the nineteenth century was followed by a second cluster in the mid-nineteenth century in larger cities like Bristol, Liverpool, Manchester and Leicester. As the fashion spread, so smaller towns across the whole of the country struggled to establish their own societies. Indeed, Allen (2000a) records that approximately 1000 scientific societies existed at some point during the nineteenth century, roughly 250 of which left published *Transactions*. Those societies in operation in the 1860s and 1870s were

witness to increasing levels of organization, becoming 'richer, more mobile and better documented, routinely keeping records and ensuring they survived' (Alberti 2001, 120).

The spaces of natural history

Some of the earliest natural history societies were peripatetic by nature. The Berwickshire Club (established in 1831), for instance, had no formal meeting rooms, it being a purely field-based society. Members would meet in a designated district, with an extended period of field study sandwiched between meals at the local inn (Allen 1994; Withers 2001). In similar manner to the Berwickshire Club, the various artisan botanical societies of Lancashire routinely used pubs as meeting places and repositories for their libraries, paying a 'wet rent' – the consumption of satisfactory quantities of beer – to the publican in return for the provision of rooms and the safe-keeping of funds, books and collections (Secord 1994a). This practice was viewed in a dim light by middle-class social reformers and Sabbatarians. As Secord notes, the discovery

that rational pursuits, especially science, were associated with the pub was inconceivable to many in the middle class: it trumpeted the failure of such knowledge to imbue its consumers with the desired notions of order, self-control and sobriety. (Secord 1994a, 274; see also Kneale 1999 2001)

Although the spaces of artisan natural history were sedentary and firmly delimited (unlike the peripatetic geographies of the Berwickshire Club), the pub's immoral associations with the evils of alcohol meant that those societies' activities could not necessarily be judged alongside the efforts of other scientific societies that chose more salubrious localities for their discussions. Many artisan natural history societies were actually persuaded to desist from using public houses as meeting places, with members choosing instead to congregate in school halls or a member's home.

In contrast to the artisan naturalists of Lancashire, Britain's many middle-class scientific societies of the later nineteenth century were trenchant in their belief that any such morally upright organization should have, or at least rent, its own property. According to Allen,

status was founded on property – and a propertyless body, it was assumed, must be no less contemptible and ineffective than a propertyless man or woman. (Allen 1994, 145)

It is for this reason that peripatetic natural history organizations like the Berwickshire Club were the exception (even if, given the lack of rent, they tended to do rather better financially than their building-bound contemporaries) and why so many other societies chose to locate their activities within designated spaces. Indeed, the prevalence of the belief that such societies and their paraphernalia should have a permanent home, along with well-defined spatial parameters for particular activities within and beyond them, is evidenced in the extraordinary ubiquity of these organizations' basic features.

It was common for nineteenth-century scientific societies to have a formal debating-chamber where meetings might be held and papers presented (Allen 2000b). A museum was another crucial space for these groups, housing the society's collections of natural objects, antiquarian relics, maps, models and other curiosities, as well as its library. Allen notes that natural history societies in Newcastle, York and Torquay actually developed out of those towns' museums and libraries (Allen 2000b), whilst other societies, whose museums were proving to be something of a financial burden, made use of the *Free Libraries and Museums Act* of 1850, whereby a yearly rate of one halfpenny in the pound could be imposed upon inhabitants of towns with a population in excess of 10 000 to defray the necessary expenses of establishing or running such an institution (Forgan 1994). The last space, of course – external to these others, but nonetheless defined by them – was the field site. All, or a combination of all, these sites could be found in societies as far apart as Newcastle, Bristol and Truro (Kargon 1977; Inkster and Morrell 1983; Allen 2000b).

A tradition of correspondence in the natural sciences fed the transmission of this ideal, spatial, model (Secord 1994b). It was private correspondence that sustained British natural history during a widespread lull in the early eighteenth century (Rusnock 1999) – all the more significant when we take into consideration the costs of postage at that time – and this continued to function as a scientific principle through the nineteenth century (Allen 1994). Secord has argued that the assiduous, even compulsive, practice of exchanging information and specimens between naturalists and their societies highlights the discipline's dependence on a gift-exchange economy (Secord 1994b). Relying on the 'elaborate etiquette' of 'reciprocal obligations', this economy 'allowed the emergence of

networks because it enabled one to know who to trust' (Secord 1994b, 389, 393). Members of regional clubs literally felt obliged to respond to all enquiries and requests from other groups elsewhere in the country and these exchanges, bolstered by the knowledge that whom one corresponded with was from a similar social stratum, resulted in a remarkable similarity between natural history societies across what were, at that time, still quite considerable distances. Members also regularly moved amongst clubs, able to continue to pursue their interests in a new setting if they had been required to move for work or health, with the immediate familiarity of the new society's organizational structure making this transition much easier.

Despite the close organizational parallels between societies across the country, many justified their existence by laying claim to regional distinctiveness. Butlin has argued that many county societies were 'symbolic of a fierce and militant localism that was determined to resist the erosion of distinctive regional and local differences in culture and tradition' (Butlin 1990, 242). Whether this was the case for regional natural history societies is debatable. It is certainly true that they believed it their duty to promote better understandings of the particularities of their local natural and cultural history, to their members, to the local public more generally, as well as to other scientists around the country and wider world. In their shaping of local people into rigorous, methodical and collective observers of their own region, in the production of a sense of pride in the local area, and in the provision of accurate, useful and comparable information on the region to researchers elsewhere in the world, so regional natural history societies found justification for their existence.

The paper moves on now to consider one county and society in particular, to West Cornwall and the Penzance Natural History and Antiquarian Society (hereafter PNHAS). The next section deals with each of the Society's three main scientific spaces in turn – its museum, its lecture hall and its field excursions. Before all this, the PNHAS is placed within its broader regional and scientific context.

Institutionalizing science in nineteenth-century Cornwall

Richard Carew's *Survey of Cornwall*, published in 1602, was perhaps the first attempt to develop a systematic inquiry into the Cornish region; indeed,

Carew's volume has been cited as a good example of the impact of early modern chorography on the development of regional 'personalities' (Jankovic 2000a, 80). Many of Carew's observations were employed by seventeenth to nineteenth-century natural historians, either visiting or residing in the region, John Ray, Edward Lhwyd, William Borlase and John Stackhouse amongst them (Turk 1959; Butlin 1990). These associations highlight the close affinities that existed between eighteenth-century natural history and antiquarianism and seventeenth-century chorographical inquiry, if only in their close attention to place. Butlin (1990) has argued that early modern natural history and antiquarianism should be included alongside chorography as important early shapers of regional distinctiveness. Indeed, and following Jankovic (2000b, 11), we might argue that natural history and antiquarianism during the long eighteenth century were simply chorography by another name, in that for all, regional study – what Jankovic (2000b, 11) terms an investigation into 'the meaning of locality' – was 'an exclusive end of investigation'.

Nineteenth-century Cornish natural history diverged from these earlier inquiries in a number of ways. First, the county's scientific practitioners began to see their region as less a locality to be studied for its own sake, in the manner of, say, Borlase, and more as 'a specimen in a larger entity' (Jankovic 2000b, 11). Whilst this is not to say that Cornish natural historians became uninterested in their own locality, it is to argue that they became aware that any such investigation should contribute to endeavours at a much larger scale – the distribution of species for instance, or the identification of the laws of nature. (We will come back to this theme again later in the paper.)

Secondly, Cornish science became a group endeavour, its use of a set of institutional spaces for *collective* and *associational* scientific inquiry marking one clear division from earlier, individual, approaches to regional study. In 1814 the Royal Geological Society of Cornwall (RGSC) was founded in Penzance, the title bestowed by patronage from the then Prince Regent (Turk 1959). At the outset the Society was utilitarian rather than philosophical, pulling in local members with interests in the development of the region's mining (Turk 1959; see also Morrell 1983). In 1815, a Museum was formed – the first scientific museum ever established in the county – whilst in 1821 a

new class of Honorary Member was inaugurated so that the Society could assure for itself contacts and recognition from across Britain and the wider world.

In 1818, the second of the Cornish Royal Societies was formed – the Royal Institution of Cornwall (RIC), located in Truro. Based on the Royal Institution of Great Britain, and initially entitled the Cornwall Philosophical Institution, the RIC was particularly interested in the applied sciences, especially geology, mineralogy, engineering and agriculture (Roderick and Stephens 1972). Whilst most of the ordinary members of the RIC were of the middle classes, it was led by the local landed gentry, including the likes of Lord Exmouth, Sir Charles Lemon, Lord St Levan and the Earl of Mount Edgcumbe (Roderick and Stephens 1972). Meanwhile, the third of the Royal Societies – the Royal Cornwall Polytechnic Society – was formed in Falmouth in 1833, financed by the local Fox family. Again, this Society began its life as a utilitarian society, promoting scientific and mechanical inventions and improvements, as well as the education of the Cornish mineworker (see Kargon (1977) on similar developments in nineteenth-century Manchester). The local gentry, such as Lord de Dunstanville and Sir Charles Lemon, again played key roles in the organization of the Society.

In 1833, a fourth Cornish Royal Society was formed in Truro – the Royal Horticultural Society of Cornwall (RHCS). Although it went into decline in 1842, the Society marked a shift in emphasis from the aims of the previously mentioned institutions, making explicit its promotion of 'a well ordered life' and the 'superior gratifications of rational and useful pursuits' (Anon 1833, 7). Whilst all of the societies mentioned above contributed to the construction of a civil industrial society, the RHCS was the first to mark out moral advancement as an aim in itself rather than as a fortunate by-product of industrial innovation. The Society was also the first in the region to promote science as a recreational activity rather than as an aid to working life. In this sense, we might see the RHCS as an important precursor to the PNHAS. Although the PNHAS was the last of Cornwall's nineteenth-century natural history societies to be established – it was founded in Penzance in 1839 – the Society was the first to be devoted entirely to the pursuit of science as a solely recreational activity.

The PNHAS drew the bulk of its membership from the population living in or around the town of Penzance, although some came from the wider West Cornwall area and a very small number from towns in the west of England more generally, the university towns of Oxford and Cambridge, from London and from the British colonies of Australia and New Zealand. In 1880, the price of ordinary yearly membership was set at a half guinea and it cost five guineas for life membership (Ross 1880a). Those wishing to become Ordinary Members had to be recommended by two current members of the Society, with these recommendations placed in some conspicuous part of the museum seven days prior to the date of meeting at which the election was proposed to take place, whilst candidates could be excluded by a majority of the committee voting against them, or a quarter of those voting if the election was held at a General Meeting (Roderick and Stephens 1972).

Whilst there is no explicit indication of the social groups to which these individuals belonged, the costs of joining the Society, as well as the rather incestuous nomination procedure, would have mitigated against anyone other than the middle classes and local gentry from becoming members. It was certainly the case that the more prominent PNHAS members were exclusively from the upper middle class, holders of significant occupational positions in Penzance or around and involved in medicine, the law, politics, town administration, banking or the church. It is worth noting that the PNHAS, unlike Cornwall's various Royal societies, lacked any involvement from the local gentry and relied instead on this other social group to organize the Society's activities and accounts.

Despite its industrial significance, which did of course enter into a terminal decline in the second half of the nineteenth century, Cornwall was never graced with a university of its own. As such, the organizations discussed above were the only outlets for Cornishmen and women wishing to develop their interests in science in a communal atmosphere, whether they were novices or leaders in their field. It is with these groups that we should begin any enquiry into the conduct of science in nineteenth-century Cornwall. The paper moves on now to consider in detail the operations of one Cornish scientific society in particular, the PNHAS.

The Penzance Natural History and Antiquarian Society

From the early 1840s, the Society had established its museum in the circular room under the dome of Penzance's Market House, which stood at the head of the town's main commercial street, Market Jew Street (Millett 1880). However, in 1867 work was completed on a large complex, also in the centre of the town, named the Public Buildings (Plate 1). Along with other societies, public institutions and administrative bodies, the PNHAS re-located there. Its museum was positioned in the central portion of the building, next to the Public Library, whilst the RGSC housed its museum and library in the west wing, with the Town Hall, Police Office and the offices of the municipal body in the east wing (Corporation of Penzance 1887).

Since its inception, the Society went through periods of activity and inactivity, falling dormant in 1856, only to be revived in 1862 so that the Cambrian Archaeological Society could be met by a comparable body on their visit to Penzance in that same year. There was a second decline from 1872 to 1879, blamed on the financial burden of moving the museum to its new residence and the resulting debt, as well as on a collapse of local support of the annual excursions (Millett 1880). However, with the help of the then mayor of Penzance, Charles Campbell Ross, the PNHAS was revived once more.

In 1880 the Society raised money through donations and so re-acquired the collections of the PNHAS museum – the Public Buildings Company had previously held the contents of the museum in defray for the rent owed to the Company by the Society (Ross 1880b 1880c). Further work was carried out that year in terms of the defining and achievement of the aims of the Society. Although they underwent some revision, the original objectives of the PNHAS – ‘the cultivation of the science of Natural History, and . . . the investigation of the Antiquities referring to the early inhabitants of our county’ (Anon 1845, ii) – continued as its foundational principles.² The Society also began compiling the third volume of its own *Transactions*, two volumes, covering 1845–50 and 1851–5, having already been published during the last substantive period of activity. As we shall see, members also went about cleaning and re-ordering the museum (Millett 1880; Willan 1880), and re-constituting a series of field excursions and monthly evening

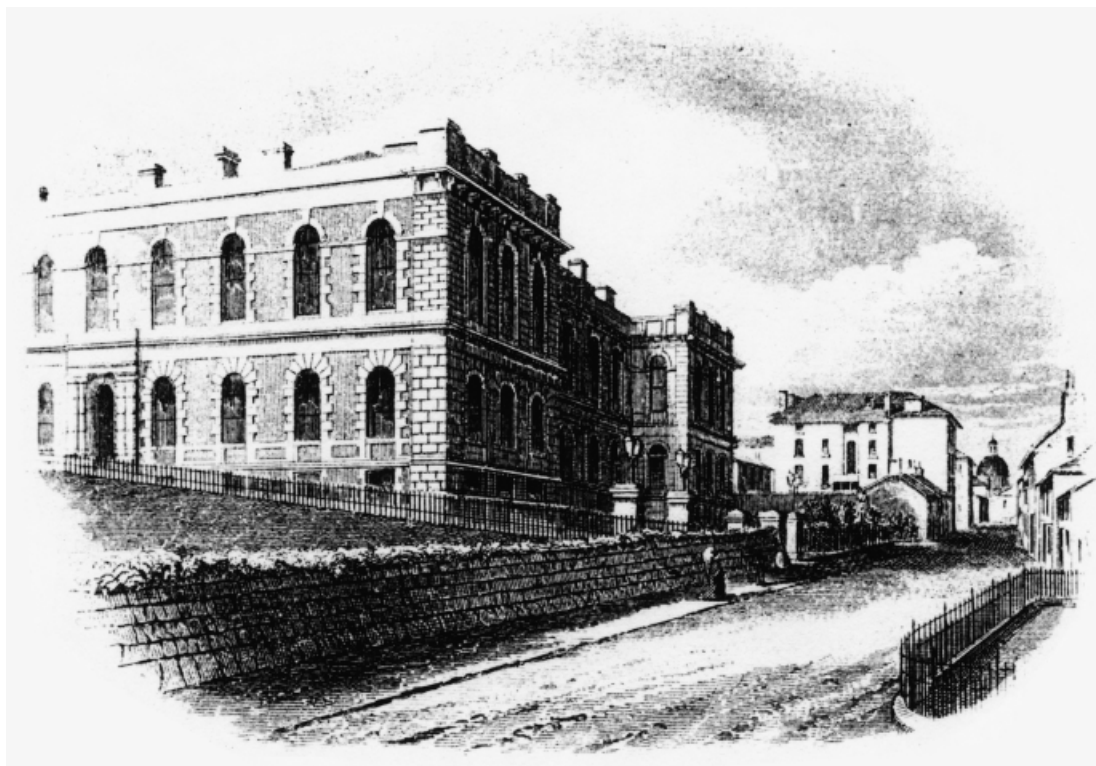


Plate 1 The Public Buildings, Penzance

'conversaciones'. The maintenance of these endeavours preoccupied the PNHAS for the remainder of its natural life up until the outbreak of the Second World War. Like the many other natural history and antiquarian societies across Britain in the nineteenth and early twentieth centuries, these spaces were central to the promotion of the PNHAS's aims. Perhaps the most important of those spaces was the museum; at the very least it was around the museum that the PNHAS's various activities revolved.

Spacing the PNHAS

The Museum

There is a resistless influence, silently recurring around us, felt in the Hamlet as in the Town, beside the Cottage hearth, as in the home of luxury – that of intellectual power. ... Let it's [sic] exercise be directed to the [Museum] acquisitions here offered, which will lead to a greater refinement in Manners, Morals, and Taste; and a spirit of research, and observation, whose resources cannot be dried up. (Anon 1845, iii)

In a lengthy inaugural address to Penzance's newly formed Natural History Society (the PNHAS's original title) – preserved, unsigned, in the Society's first minute book (although presumably delivered by the incumbent president Joseph Carne) and quoted above – the speaker went about justifying the Society's existence. Pointing to the rich natural environment of the region, and to the presence there of many who were knowledgeable of the natural sciences, he also noted that 'many others, for want of the needful facilities and helps, have been discouraged, and have made but little progress'.

To facilitate the researches of the former group and to improve the lot of the latter, the speaker proposed the establishment of a museum, which would be administered by, and form the nucleus of, the new Society. However, the museum was not simply to serve to increase regional and general scientific knowledge, it was also to improve the character and integrity of its users. The construction of these geographies, knowledges and moral subjects relied on the provision of a prominent

display space that was organized to best situate and explain the local natural and cultural objects of educational and scientific interest. This was by no means an uncontested enterprise. At a time when some metropolitan museums were developing their collections towards either mass education or specialist research establishments (see Yanni 1999), provincial museums like this one were left to attempt to accommodate an element of both; indeed this and other similar societies believed it was their moral duty to do so (Forgan 1994). How the museum should have been laid out, how its collections should have been grouped and to whom it was catering were the vital questions that preoccupied the Society in the latter years of the nineteenth century.

The museum's display cabinets were accumulated over a period of years, paid for through subscriptions, fund-raising events, donations and, in one case, the sale of a valuable edition of John Gould's *Birds of Europe* – in a personal letter to Thomas Cornish, John Boase (1869) urged the sale of the book so as to get the Society out of debt, and to buy new cabinets and a number of books that might form 'the nucleus of a much needed library'. While some of the collections were the result of proactive purchases, most were obtained through donation by members. This ad hoc and unpredictable process of accumulation resulted in a somewhat random collection of objects, with a glut of some specimens and a dearth of others. Donations from members – often motivated by an early ruling that allowed donors free access to the museum (Anon 1839) – ranged from collections of Brazilian birds to a set of local star-fish and sea urchins, from papier-mâché models of ancient megaliths to rare pieces of metalwork (Plate 2). Putting these miscellaneous objects in order was, then, something of a headache for the curators, especially given the belief that only a 'systematic arrangement' of the museum's collections would render it either educational or scientific (Tregelles 1890, 198). These anxieties were certainly not confined to Cornwall – Alberti (2001) has highlighted similar concerns in some of Yorkshire's various Philosophical Societies. These acquisitional and organizational anxieties were part of a generally held dilemma over the objectives of provincial museums: should local museums, whether they be in Penzance or Leeds, attempt to provide a general picture of the world's natural history, including examples of spectacular and

rare objects, or an exhaustive collection of the region's own flora and fauna?

A guide to the region and a book of nature

The PNHAS was emphatic about the greater importance of the presentation of a complete display of *local* natural history. Nonetheless, the curators did not decide to remove the museum's miscellaneous objects from further afield, choosing instead to organize its museum by *separating* local and foreign specimens, the latter distinguished by 'the placing in one corner of each case a small disc of brightly-coloured paper' (Anon 1886, 210). The larger, more colourful, and strange specimens 'from the tropics or distant lands', it was felt, would have so dominated their Cornish equivalents that the latter would have been 'practically lost to view' (Lach-Syzrma 1889b, 223–4).³ It is likely that the curators would have jettisoned these exotic objects from their displays altogether if they could, but were prevented from doing so by the fact that donations of these specimens far outstripped those of a more indigenous variety, and that these objects proved more of a draw to potential visitors. Their presence was justified retrospectively as having 'merely an educational value' (Lach-Syzrma 1889b, 224). Meanwhile, constant exhortations went out to members, imploring them to collect and donate local specimens.

The Society based their emphasis on Cornish specimens upon the argument that local museums should always function as a map or guidebook to the surrounding region. 'Scientific men and specialists', one retiring president suggested:

like to see the flora and fauna of the places they visit. It gives them a fresh interest in their travels. Also local students may like to know what specimens are to be found, and have been found, in their neighbourhood. (Lach-Syzrma 1889b, 224)

The PNHAS museum was, then, meant to act as a metonym for the west Cornwall area, an entry-point into the region's natural history for those serious travellers intending to continue on into the thing itself. Beyond these metaphors, there was also a literal aspect to the claim. With the laying of the regulation broad-gauge track between Truro and Penzance in 1867, the town was, for the first time, linked directly to London Paddington station (Bennett 1988; see also Freeman 1999). Whilst Penzance station served as the gateway to the rest of Britain for the inhabitants of the extreme



Plate 2 'Ancient Urns &c. Found Near Penzance'. Objects exhibited in the PNHAS museum and reprinted in the PNHAS Transactions

south-west, it became, in the last decades of the nineteenth century, far more important as the point of arrival for the many Victorian tourists choosing to vacation there.

More than simply deciding to divide the museum into foreign and local, how these collections of Cornish objects could be best shown off was also debated. In particular, prominent members of the PNHAS favoured detailed descriptions over unadorned specimens. Without descriptions of the objects on display there was a danger that students and tourists might pass them over unnoticed because they were not understood. To combat this 'common evil', a full description was recommended, written on a card by the specimen's side. In such manner, the museum would not only be a guidebook to the local region beyond its walls, but also a book of natural history itself, 'with, instead of pictured illustrations, the actual specimens themselves' (Lach-Syzrma 1889b, 224). These labels were to be written in both Latin and English and so made yet more accessible (Lach-Syzrma 1890). General, well-labelled collections of typical specimens, as Alberti has noted in relation to museologists in Yorkshire, were required so as to teach those interested in natural history 'something of the elements of their science' (Alberti 2001, 131).

Although the annotation of collections might seem an obvious suggestion today, commentaries on museum specimens only became widespread in the 1870s, in the wake of attempts to open out access to collections to a wider public beyond the sphere of the privileged gentleman scientist (Forgan 1994). Certainly for the PNHAS such a schema was part of its plan to increase the museum's educational 'value', complying with broader trends to promote 'museological self-improvement' amongst the leisured classes as well as the labouring classes, women and children (Forgan 1994, 145). This endeavour was pronounced to be a success by the curators, claiming that 200 visitors had inspected the collections over the year 1887–8 (Symons *et al.* 1888, 10). However, as a counter-point, in 1891, they complained that local residents and even PNHAS members 'seem rather to ignore, or at any rate to evince no practical interest in, the collection' (Baily *et al.* 1891, 296). This problem might be situated within a widely held contemporary belief that museums themselves 'could never be as beneficial a pedagogic tool as the field or the lab' (Alberti 2001, 130–1), or for that matter other spaces of instruction and

display. Whilst prominent members of the PNHAS tended to cleave strongly to the belief that the museum was still a more central institution than the laboratory in the conduct of Victorian science (Yanni 1999), it was nonetheless the case that for them the museum could only work to best effect in conjunction with other spaces. This brings us on to the second of the Society's sites of scientific work – the *conversazione* – where most of the Society's presentations, discussions and, to a lesser extent, experimental work were carried out.

The conversazione

Tea and coffee having been served in the library, the company assembled in the dining-room, where the President on behalf of himself and Mrs. Ross formally offered a hearty welcome to the Society, explaining that one reason why they had invited the members to meet there was that ... he thought it wise to interest all by making their meetings as social as possible. [...] The President then called on MR T. CORNISH, who after explaining that these monthly meetings are designed to give opportunities for the exhibition of, and a free chat about, matters of no very solemn import, proceeded to exhibit some articles that had been recently added to his museum. (Anon 1881, no page)

After Mr Cornish's presentation, at the PNHAS's inaugural evening *conversazione* of the 1881 winter season, the members moved through to the drawing room to listen to music, and then back to the library, where supper had been laid.

By the third of the winter gatherings, all this had changed, with the members gathering instead in the more austere setting of the museum of the Geological Society: Mr Cornish, taking the chair in the absence of the President,

entered his protest against the very pleasant practice of holding these meetings at the houses of their presidents. The evenings spent at the residences of Mr Borlase and Mr Ross were evenings of extreme enjoyment, but hardly consistent with the business of the Society, and he suggested in future that the meetings be held in the museum and nowhere else' (Anon. 1881, no page)

From then on the Society met regularly in the RGSC museum for its committee and public meetings, and its *conversazioni*. This changed again in 1895, when the Society installed gas-lamps in its own museum, so rendering the room usable in the winter evenings. This enabled members to give their papers in front of the specimens, species

and objects to which they referred, making their presentations 'more instructive and interesting' (Anon undated).

These pragmatic shifts in the Society's practices and procedures regarding their lecture series are, like those of the museum, illustrative of more fundamental concerns about the social, educational and scientific roles held by the PNHAS in particular, and other scientific societies more generally. For instance, should science be conducted in private spaces or in the public realm? Should scientific meetings be formal, scholarly, and austere, or more informal, sociable and entertaining? Should they be aimed only at the serious-minded and informed expert, or should they attempt to educate a broader audience, including women and children? And finally, with these previous queries in mind, how should the meetings themselves be organized, conducted and accompanied?

Public science as private sociability

The holding of regular meetings for the presentation of scientific papers grew out of two inter-related traditions: the gathering of members immediately after field excursions so as to continue the discussions of the day; and the desire to provide an alternative meeting place for members in the winter months, when outdoor gatherings were rendered impractical by bad weather and short daylight hours. The meetings, and the papers therein, were publicized to members and friends through the local press and a series of hand-drawn posters (Plate 3). Although the subject of many of these posters were related closely to serious natural history topics – pictures of flowers, landscapes or insects for instance – others were meant to be simply comical. Indeed, their tone was obviously meant to communicate a light-hearted side to the Society's more serious ambitions; an attitude that Allen (1994 2000a) identifies as widespread amongst natural history societies operating at the time.

The papers themselves were diverse in the extreme, although most followed the principle that they should throw 'some light on scientific subjects having relation to the neighbourhood' (Anon 1890, 271). Although the majority were concerned with natural history subjects, there was a significant corpus that concentrated on, amongst other things, antiquarianism, local history, folklore, religious architecture, and dialect and language. Many of

these papers went on to be published in the PNHAS *Transactions*.

There were, on occasion, papers of a more fanciful, comical or satirical nature. Members presented ghost stories for instance, another amused the audience with a recitation of past scientific blunders, whilst another, Mr Millett, spoke to the title, 'The Ascent of Man', which

led most of the audience to anticipate a philosophical treatise demolishing the Darwinian hypothesis. The surprise therefore was great and general when, instead of this the author commenced reading his own most amusing revised version of the familiar nursery tale, 'Jack and the Beanstalk'. (Anon 1885a, 100)⁴

Despite Mr Cornish's admonition of the frivolities of early meetings, occasional conversaciones also included some form of entertainment, usually the performance of a piece of music by one of the 'Lady' members, or wife or daughter of one of the male members. It would be unfair to imply that this was the only role granted to women by the PNHAS. For instance, in 1891, the President W. S. Bennett, appealed for more women to 'do their share' in the presentation of papers at the meetings (Anon 1892, 387). However, that only a few women ever submitted anything for presentation is patently unsurprising, given the very small number of females belonging to the Society and the very masculine nature of the debates (Allen 1994; Shteir 1996).

The PNHAS believed that their scientific lectures and discussions should take place in an appropriate *public* space – first the museum of the RGSC and then in their own exhibition room – even if the close-knit nature of the Society continued to foster a 'private sociability' during the meetings themselves (Withers 1998, 49). Moreover, as Cornish's remarks – quoted above – make clear, the PNHAS conversaciones became increasingly formal, organized and serious-minded, with less room for the more 'indulgent' of members' interests. However, in similar manner to its intentions over the display of foreign acquisitions, it was not the case that the less 'scientific' aspects of the Society's meetings were banished from its calendar altogether. The provision of meals certainly stopped, although the interspersing of scientific papers with musical interludes, performed by women members, was employed on occasion. The inclusion of more light-hearted papers into the programme of lectures was, if not encouraged, certainly tolerated and



Plate 3 A poster advertising the PNHAS's February meeting of 1883

Source: by courtesy of the Morrab Library, Penzance

enjoyed. These changes, along with the use of new display technologies⁵ and attempts to widen participation in the lectures, signalled an increasing concern to use the lectures as a space for educational as well as scientific improvement (see Allen 2000b for a similar argument). Comical posters, amusing papers and spectacular exhibitions were meant to draw in larger audiences so as to address the former concern, whilst the presentation of papers in front of museum exhibits was another instance of the Society's attempts to construct a regional and natural history through its key spaces. It is possible to identify similar tenets running through the Society's field excursions.

The field excursion

It is now a characteristic of our Society that we take the position of a field club as well as a scientific society for the consideration of papers; that we continue our monthly meetings through the summer as well as in winter. The value of this system must be manifest to all who have taken part in it. 1. It keeps our Society together all the year round. 2. It enables our members to study in situ our local curiosities – artificial or natural. 3. It diffuses knowledge of our environs among our members. 4. It shows, through the Press, to strangers and outsiders the treasures in an antiquarian or natural sense that we possess. 5. It combines healthy recreation with head-work. 6. It extends our sphere of usefulness. (Lach-Syzrma 1889b, 220)

Lach-Syzrma's paper, quoted above, illuminates the range of purposes that the Society felt its field programme could address for its members. However, beyond the provision of physical exertion and 'head-work', these group forays into the countryside surrounding Penzance were most generally concerned with the provision of an education in the local environment for both locals and visitors. The excursions were part of the Society's programme of regional re-inscription – re-writing west Cornwall as a book of natural and cultural history (on the histories of fieldwork more generally, see Kuklick and Kohler 1996; Allen 2000a).

Each of the day-long excursions would take in quite a number of sites, with either prominent members of the Society giving lectures on the locations visited, or other relevant parties being invited to do so (Plate 4). It was common, for instance, for vicars to be asked to give lectures on their own church if visited as part of the trip. Indeed, the rather formal attire of excursionists, in Plate 4, was certainly desirable, and even a neces-

sity, when visits to churches and stately homes were part of the itinerary.

Movements and refreshments

Brakes – horse-drawn carriages – were used regularly to move the excursionists around and these were gradually, although only partially, replaced by the train and ferry.⁶ The use of carriages suited the peripatetic nature of the day-long excursions, with numerous visits to sites of interest made along a pre-determined route. Nonetheless, trains were ideal for the monthly afternoon-only excursions, initiated in 1888, where trips were often only made to one location. Train travel to excursion sites became more common for the PNHAS in the 1890s.

To get to sites of interest that were beyond the reach of either brakes or trains, the excursionists were routinely expected to do some walking. These were described in turn as 'hikes', 'trudges', 'processions' and 'climbs' and set into relief against descriptions of scenic landscapes observed, at rest, through the carriage window. The rather cumbersome nature of the excursionists' clothing – especially the women's dresses – go some way to explain the reticence of participants to venture beyond the realms of their transportation, especially when they were required to cross relatively inhospitable, and certainly muddy and wet, moorland.

While these moments of exertion complimented the general fervour of the day, the Society took great care to ensure that excursionists were granted plenty of time for lunch and afternoon tea. This interjection of a more leisurely pace into the hurry of the day was encouraged by Lach-Syzrma, during his presidency in 1884–5:

It is a good thing in excursions of this character not to be too hurried. It is better both from a scientific standpoint to have free discussion, while calm consideration of the curiosities is desirable; and also from the point of comfort and enjoyment. (Lach-Syzrma 1885, 131–2)

Lach-Syzrma went on to criticize the 'danger of hurry' that he claimed dogged the London societies, and recommended instead to extend the length of the day. Field research, he said, 'requires a little time, and it is more agreeable not to be hurried' (Lach-Syzrma 1885, 131–2).

Cornish's suggestion that the excursions become more leisurely obviously carried with it a danger that the days became nothing more than thinly veiled pleasure trips. Certainly it was felt that



Plate 4 Excursionists during a visit to St Buryan Church, Cornwall, on the 28 August 1885. Note the large number of women present in the party

Source: by courtesy of the Morrab Library, Penzance

during a visit to Chywoon Castle in 1891 'the true antiquarian spirit was lacking in some members . . . for from more than one was the irreverent exclamation heard, "What a splendid place for a picnic!"' (Anon 1891, 308–9). This was not a 'problem' that dogged only the PNHAS: Allen notes that the excursions of many scientific societies around Britain in the later nineteenth century began to accrue the attributes of the pleasure-trip rather than of scholarly exploration (Allen 1994). He goes on to argue that the afternoon-only excursion was the result of this shift, and was aimed at catering for the increasing numbers of women desirous to attend, a hypothesis certainly born out in Plate 4.

Nonetheless, the pursuit of pleasure during excursions was not *entirely* detrimental to the aims of the Society, and was seen as an opportunity for education into the local region. The journeys between the various locations on the itinerary yielded a variety of vistas and the Society took these as a chance to instruct its members in landscape appreciation and the local topography:

When the crown of the hill was reached, the beautiful panorama for some time absorbed the attention of all. The hill perhaps commands as grand a view as can be seen in England. On one side lies Mount's Bay, on the other St. Ives Bay, the coast trending away as far as Trevoise Head, whilst looking up the country the horizon is bounded by the bold outline of St. Agnes Beacon, Carn Brea, Carn Marth, and the Tregoning and Godolphin hills. (Anon 1883, 195)

Matless has termed this an 'outlook geography', a culture of looking that places emphasis on observation from a high point (Matless 1992). This outlook geography, prevalent in activities popularized in the early twentieth century, interwove particular forms of movement through the landscape with forms of visual appreciation and moral virtue. For the PNHAS, these 'lessons' in landscape were by no means removed from or fleeting adjuncts to the real work of the day. Rather, the excursions were effective ways of encouraging appreciation for local landscapes and, in particular, a forum for the promotion of preservation of local sites of

scientific interest, threatened by building schemes or maulings by farmers. In the next and final substantive section, we move on to consider in more detail the ways in which the Society went about constructing understandings of the region and appropriate ways of approaching its study.

Writing the region

It is now possible to reconsider Withers' claim, cited at the beginning of the paper, that Victorian natural history societies played significant roles in the construction of local and national scientific knowledges. In the context of the work of the PNHAS, we can identify three significant ways in which the Society and its museum, lectures and excursions worked towards these ends: by encouraging the labours of local natural historians; by constructing West Cornwall as a site of singular scientific importance; and by defining an agenda for regional scientific study.

First, the PNHAS provided both discreet and networked spaces for the conduct of a variety of scientific work by its members. The field excursions gave the opportunity for assisted research into the natural and historical objects of interest in West Cornwall, whilst the *conversazioni* were a way of continuing group discussion and individual presentations into the evening after a day's fieldwork, and into the winter months when excursions were impractical. The museum was also connected intimately to these two other spaces: as a repository for objects collected during fieldwork, and as a site for the staging of evening events. In an area lacking the amenities of Britain's larger towns and cities, the PNHAS therefore provided an important set of spaces for collaboration amongst 'the little knot of scientific men and, I may add . . . ladies in Cornwall, who have kept our work going' (Lach-Syzrma 1889b, 217). Although Truro's Royal Institution of Cornwall became increasingly concerned with natural history and antiquarianism in the later nineteenth century, members of the two towns' societies tended to maintain close allegiances to their own club rather than explore the possibilities of collaboration (cf. Butlin 1990). As Lach-Syzrma put it in 1885, the PNHAS

has earned a name and position in the scientific world, and though schemes for amalgamation with others may have their attractive sides, yet the wisest thing is for our westernmost borough to keep up its society, of which it may well be proud. (Lach-Syzrma 1885, 132)

PNHAS members were as loyal to their region as they were to their society. The PNHAS regularly justified its existence through direct reference to West Cornwall's supposedly unique wildlife and heritage. The PNHAS, it was suggested, had a 'more promising field to work upon than the societies of most of the small provincial towns' throughout the rest of England (Lach-Syzrma 1889b, 217), due, for instance, to its particularly mild climate, its proximity to the sea and its wealth of pre-historic remains (Lach-Syzrma 1889a, 131). This was a view reinforced routinely during fieldwork by excursion guides, and substantiated, as we have seen, through the Museum exhibits and the papers presented at the monthly *conversazioni*.

It was commonplace for speakers to reiterate the natural uniqueness and diversity of the region in relation to the possibilities offered elsewhere in the wider world. For instance, during one fieldtrip to the Lizard Peninsula in 1890, the excursion leader argued that the geological and botanical diversity of West Cornwall was greater even than whole swaths of continental Europe:

Many of the midland counties of England are not so distinct from each other as [the Cornish district of] Kerrier is from Penwith, and one might almost add that even parts of Germany and Belgium are more like each other, and also of Germany and France, than these two regions. (Anon 1890, 212)

From the first then, the PNHAS took it upon itself to 'direct the attention of our members to the treasures of natural history which exist almost at our own doors' (Carne 1850, 360).

Taxonomizing the region

Whilst the institutional sites of the PNHAS allowed individuals to present their own local studies, these spaces were perhaps most significant in their promotion of a local study *agenda*. In other words, these sites were training grounds for natural history, teaching potential field-researchers how to correctly and scientifically observe, appreciate, interact with and record their region (cf. Matless 1999). This sentiment was most evident in the Society's exhortations to catalogue indigenous species and compose species inventories.

For the PNHAS, the production of regional taxonomies was of paramount importance, for two main reasons. First, catalogues of indigenous species enhanced the writing of a regional natural

history. 'Systematic lists' would supply an 'exhaustive work' on Cornwall's flora and fauna (Millett and Marquand 1886, 219), to be embodied in the Museum displays as well as the PNHAS *Transactions*. In turn, these exhibits and publications would act as an 'index to all the natural productions of our neighbourhood' (Millett *et al.* 1850, 364) – useful records of past labours and 'an incentive to future efforts' (Millett 1881, 85). Whilst they were valued in their own right, these catalogues would also contribute to research into species' habits, growth and mutual relations (Tregelles 1887, 7) and also to national taxonomic endeavours. As John Ralfs put it in his Presidential Address, 'On the History of Botany in Cornwall':

In conclusion, I would impress upon our members the high scientific value of carefully and exhaustively noting the flora of a district, however small; for by that means alone can a clear and accurate estimate be formed of the distribution of plants in Cornwall as compared with the rest of the kingdom. (Ralfs 1885, 20)

The production of regional taxonomies was important for a second main reason. Whilst the cataloguing of West Cornwall's flora and fauna enabled the PNHAS to contribute to national debates over the prevalence and distribution of species (cf. Allen 1994), so too did it facilitate the training of scientific citizens. The twin acts of recording nature in the field and then putting those records in an order useful to others were, according to the PNHAS, the crucial skills of the local natural historian:

The great point which the Society grasped from the first, which it still maintains, and I hope always will be maintained, is the encouragement of independent research. We do not want educational lectures on divers scientific topics, still less do we want quotations from, or *resumés* of, books more or less difficult to obtain. We want facts, either those little known to the scientific world, or, better still, facts as yet unknown, *but discovered by the members*. (Lach-Syzrma 1889a, 130, emphasis added)

Lach-Syzrma and others claimed that the PNHAS had been successful in achieving these aims. It was suggested in the PNHAS *Transactions* that West Cornwall was one of the 'more favoured districts of Great Britain' for botany, but that this was less to do with the *extent* of its flora as it was 'in the degree in which its indigenous plants, from the highest to the lowest, *have been brought to light and catalogued*' (Anon 1885b, 85, emphasis added).

In other words, the success of the Society was born out, less in the voluminous nature of its records of Cornish natural history, and more in its success at educating members in the deployment of rigorous and scientifically comparable techniques of collection and recording. The rewards to be gained were certainly not incidental: for Millett, the recording of facts pertaining to either natural history or antiquarianism 'unfolds a history to the reverent and accurate observer' (Millett 1880, 11), whilst William Bolitho, taking up the mantle of President in 1885, suggested that such an observer would inevitably and 'certainly be led to look from Nature up to Nature's God' (Bolitho 1885, 203–4).

Conclusions

The PNHAS turned Cornishmen and women into natural historians, and Cornwall into a site of civic pride and a scientifically delimited space. These were achieved through and across a range of sites, from the micro-geographies of museum display cabinets up to the imagined space of national scientific endeavour. Indeed, whilst it has been the purpose of this paper to demonstrate the role local natural history societies played in the construction of regional scientific knowledges, it is important to remember that the ideas and findings of these societies did not stop at their own county boundaries. Rather, the work of local societies such as the PNHAS moved across a range of social and spatial contexts, primarily through the publication of local work in society *Transactions* or *Proceedings*, but also through inter-club and member's correspondence. The extent of the PNHAS's purchasing of other society's publications, or indeed the dissemination of its own *Transactions*, was certainly not out of the ordinary amongst societies across Britain at that time. A catalogue of books owned by and held in the PNHAS library lists a range of regional and city natural history society *Transactions* from as far afield as Marlborough, Eastbourne, Glasgow, Southport, London, Belfast, Yorkshire, Plymouth, Bristol, Nottingham, Manchester and Huddersfield. Society and museum reports from Australia, the USA, Germany and France were also held (Gates 1903).

Although this is not intended to be an argument for the prevalence of a unifying national scientific project amongst local natural history and antiquarian societies, it does gesture toward the

strength of feeling they shared regarding the education of the wider populace in the natural and cultural peculiarities of their own region, and in scientific methods and principles more generally.

The study of science through its spaces of activity and across its networks of geographies also justifies the methodological tenor of this paper. Following scientific practitioners – whether naturalists in Victorian Cornwall, chemists in Restoration London (Shapin 1994) or astronomers in twentieth-century California (Soojung-Kim Pang 1996) – through their sometimes extraordinary, but more often than not mundane, spaces – gives us access to their thoughts and deeds, otherwise denied to us (Latour 1987; Shapin 1998). Investigating science in this way also helps us to ‘recover’ the activities of a range of practitioners often invisible to us if we follow a traditional hagiographic approach to the history of science (Withers 2001), with its over-emphasis on well-known researchers at the expense of the many others who laboured and contributed much to the development of science as method and tradition (Cooter and Pumfrey 1994). Lastly, this approach, far from dwelling too long on parochial cases, or making over-blown claims on the basis of specific or limited studies, is crucial if we accept that knowledge is locally produced and situated. Indeed, it is *precisely* through such an approach that we can gain better understandings of the formation of local knowledges as they operated within networks of wider-ranging intellectual cultures.

Acknowledgements

Archival research for this paper was generously supported by grants from the Institute of Cornish Studies, University of Exeter and the British Academy, and was conducted in the Morrab Library, Penzance, the Cornwall Record Office, Truro, the Courtney Library of the Royal Institution of Cornwall, Truro and the Cornish Studies Library, Redruth. Thank you to the staff of all these institutions for their generous help and support, with particular thanks to Jan and Annabel at the Morrab. Keith and Liz were, as always, helpful and supportive during periods of fieldwork. Thank you to Jonathan Tooby for preparing the plates and to the Morrab Library for granting permission to include their images in this paper. Versions of the paper were presented at the 2001 RGS/IBG conference, Plymouth, the 2001 11th International

Conference of Historical Geographers, Quebec, and the annual meeting of the Society for the History of Natural History, Shrewsbury 2002. Thanks to the audiences at all these events for their comments and criticisms. Thanks also to Ron Martin and three anonymous referees for their invaluable comments and suggestions.

Notes

- 1 The 1815 *Apothecaries Act* reinforced the principles established by the Society of Apothecaries, outlined above. The Act dictated a five-year apprenticeship for apothecaries and demanded of them some minimal acquaintance with British plants. The result was a ‘remarkable burgeoning of field classes in botany’ in British universities at that time (Allen 1994, 95; Allen 2000b).
- 2 Due to the constraints of space, this paper will not address the antiquarian activities of the PNHAS. See Naylor (submitted) for a consideration of eighteenth- and nineteenth-century Cornish antiquarianism.
- 3 The decision to separate the two types of exhibits was taken finally at a Meeting of the Council in 1888 (Cornish 1888).
- 4 It is worth drawing attention to the tenor of this note, in that it illustrates the belief amongst provincial societies of their right and ability to engage with the theories and research of leading contemporary scientific scholars.
- 5 In the 1880s the Society began to make regular use of an oxy-hydrogen lantern, or magic lantern, and an oxy-hydrogen microscope to illustrate members’ talks. These combined with more traditional forms of visual representation like maps and hand-drawn images, as well as the performance of experiments – such as Mr A.K. Bennett’s illustration of ‘the application of electricity for practical purposes by various simple and instructive experiments’ (Anon 1885a, 100) – to produce what was a highly visual space of instruction (Withers 1998; Stafford 1996).
- 6 Historians of travel have noted the relatively slow uptake of train travel in Cornwall, due to limited services and lines (Noall 1963).

References

- Alberti S 2001 Amateurs and professionals in one county: biology and natural history in late Victorian Yorkshire *Journal of the History of Biology* 34 115–47
- Allen D E 1994 [1974] *The naturalist in Britain: a social history* Princeton University Press, New Haven
- Allen D E 2000a Walking the swards: medical education and the rise and spread of the botanical field class *Archives of Natural History* 27 335–67

- Allen D E** 2000b Natural history societies and their role in disseminating information Paper presented to the annual general meeting of the Society for the History of Natural History, London
- Anon** undated Untitled newspaper article *PNHAS Minute Books 1880–1889* PNHAS Archive, Morrab Library, Penzance no pages
- Anon** 1833 Report *Annual Report of the Committee of the Royal Horticultural Society of Cornwall* E. Heard, Truro 7–8
- Anon** 1839 Rules. Museum *PNHAS Minute Books 1839–1872* Catalogue number X755/1 Cornwall Record Office 14–15
- Anon** 1845 Preface *Transactions of the PNHAS, Vol. I. 1845–1850* F.T. Vibert, Penzance ii–vii
- Anon** 1881 Untitled, undated newspaper article *PNHAS Minute Books 1880–1889* PNHAS Archive, Morrab Library, Penzance no pages
- Anon** 1883 An account of the annual excursion of the Penzance Natural History and Antiquarian Society *Transactions of the PNHAS, Vol. I. N.S. 1880–1884* William Brendon and Son, Plymouth 195–204
- Anon** 1885a Meetings of the Society *Transactions of the PNHAS, Vol. II. N.S. 1884–1888* William Brendon and Son, Plymouth 93–110
- Anon** 1885b Flora and fauna *Transactions of the PNHAS, Vol. I. N.S. 1880–1884* William Brendon and Son, Plymouth 85–6
- Anon** 1886 Meetings of the Society *Transactions of the PNHAS, Vol. II. N.S. 1884–1888* William Brendon and Son, Plymouth 201–11
- Anon** 1890 Meetings of the Society *Transactions of the PNHAS, Vol. III. N.S. 1888–1892* William Brendon and Son, Plymouth 271–86
- Anon** 1891 Excursions *Transactions of the PNHAS, Vol. III. N.S. 1888–1892* William Brendon and Son, Plymouth 305–14
- Anon** 1892 Meetings of the Society *Transactions of the PNHAS, Vol. III. N.S. 1888–1892* William Brendon and Son, Plymouth 383–95
- Baily W E, Symons J and Montgomerie H** 1891 Report of the Curators *Transactions of the PNHAS, Vol. III. N.S. 1888–1892* William Brendon and Son, Plymouth 296–7
- Bennett A** 1988 *The Great Western Railway in West Cornwall* Kingfisher Railway Productions, Southampton
- Bennett J and Mandelbrote S** 1998 *The Garden, the Ark, the Tower, the Temple: biblical metaphors of knowledge in early modern Europe* Museum of the History of Science in association with the Bodleian Library, Oxford
- Boase J J A** 1869 Letter to Thomas Cornish dated 20 October Catalogue number X755/5 PNHAS Archives, Cornwall Record Office no pages
- Bolitho W** 1885 Presidential address *Transactions of the PNHAS, Vol. II. N.S. 1884–1888* William Brendon and Son, Plymouth 231–6
- Butlin R A** 1990 Regions in England and Wales c.1600–1914 in **Dodgshon R A and Butlin R** eds *An historical geography of England and Wales* Academic Press, London 223–54
- Carne J** 1850 President's Address *Transactions of the PNHAS, Vol. I. 1845–1850* F.T. Vibert, Penzance 359–62
- Cooter R and Pumfrey S** 1994 Separate spheres and public places: reflections on the history of science popularization and science in popular culture *History of Science* 32 237–67
- Cornish T** 1888 Meeting of the council held on 2 May *PNHAS Minute Books 1880–1889* PNHAS Archive, Morrab Library, Penzance no pages
- Corporation of Penzance** 1887 *Official guide to Penzance* Beare and Son, Penzance
- Crang M** 1998 Places of practice, and the practice of science *Environment and Planning A* 30 1971–4
- Driver F** 2000 Editorial: Field-work in geography *Transactions of the Institute of British Geographers* 25 267–8
- Forgan S** 1994 The architecture of display: museums, universities and objects in nineteenth-century Britain *History of Science* 32 139–62
- Freeman M** 1999 *Railways and the Victorian imagination* Yale University Press, Yale
- Gates J** 1903 Catalogue of books of the Penzance Natural History and Antiquarian Society Catalogue number X755/1 PNHAS Archives, Cornwall Record Office no pages
- Inkster I and Morrell J** 1983 *Metropolis and province: science in British culture, 1780–1850* Hutchinson, London
- Jankovic V** 2000a The place of nature and the nature of place: the chorographic challenge to the history of British provincial science *History of Science* 38 79–113
- Jankovic V** 2000b *Reading the skies: a cultural history of weather in England, 1660–1820* Manchester University Press, Manchester
- Jenkins A C** 1978 *The naturalists: pioneers of natural history* Hamish Hamilton, London
- Kargon R H** 1977 *Science in Victorian Manchester: enterprise and expertise* Manchester University Press, Manchester
- Kneale J** 1999 'A problem of supervision': moral geographies of the nineteenth-century British public house *Journal of Historical Geography* 25 333–48
- Kneale J** 2001 The place of drink: temperance and the public, 1856–1914 *Social and Cultural Geography* 2 43–59
- Kohlstedt S G** 1988 Curiosities and cabinets: natural history museums and education on the antebellum campus *Isis* 79 405–26
- Kuklick H and Kohler R E** eds 1996 *Science in the field* *Osiris* special issue 11
- Lach-Syzrma W S** 1885 Presidential Address *Transactions of the PNHAS, Vol. II. N.S. 1884–1888* William Brendon and Son, Plymouth 127–32
- Lach-Syzrma W S** 1889a Presidential Address *Transactions of the PNHAS, Vol. III. N.S. 1888–1892* William Brendon and Son, Plymouth 129–31

- Lach-Syzrma W S** 1889b Retiring President's Address *Transactions of the PNHAS*, Vol. III. N.S. 1888–1892 William Brendon and Son, Plymouth 217–24
- Lach-Syzrma W S** 1890 Meeting of the council held on 3 April *PNHAS Minute Books 1890–1898* PNHAS Archive, Morrab Library, Penzance no pages
- Larsen A** 1996 Equipment for the field in **Jardine N, Secord J A and Spary E C** eds *Cultures of natural history* Cambridge University Press, Cambridge 358–77
- Latour B** 1987 *Science in action* Harvard University Press, Cambridge MA
- Lawrence C and Shapin S** 1998 *Science incarnate: historical embodiments of natural knowledge* Chicago University Press, Chicago
- Livingstone D** 1995 The spaces of knowledge: contributions towards a historical geography of science *Environment and Planning D: Society and Space* 13 5–34
- Livingstone D** 2000 Making space for science *Erdkunde* 54 285–96
- Livingstone D** 2003 *Spaces of science* Chicago University Press, Chicago
- Matless D** 1992 Regional surveys and local knowledges: the geographical imagination in Britain, 1918–39 *Transactions of the Institute of British Geographers* 17 264–80
- Matless D** 1999 The uses of cartographic literacy: mapping, survey and citizenship in twentieth century Britain in **Cosgrove D** ed *Mappings* Reaktion Press, London 193–212
- Millett G B** 1880 Report *Transactions of the PNHAS*, Vol. I. N.S. 1880–1884 William Brendon and Son, Plymouth 7–11
- Millett G B** 1881 Report *Transactions of the PNHAS*, Vol. I. N.S. 1880–1884 William Brendon and Son, Plymouth 83–4
- Millett G B and Marquand E D** 1886 Report *Transactions of the PNHAS*, Vol. II. N.S. 1884–1888 William Brendon and Son, Plymouth 219–21
- Millett G N R, Rodd E H and Couch R Q** 1850 Annual report *Transactions of the PNHAS*, Vol. I. 1845–1850 F.T. Vibert, Penzance 363–7
- Morrell J** 1983 Economic and ornamental geology: the Geological and Polytechnic Society of the West Riding of Yorkshire, 1837–53 in **Inkster I and Morrell J** 1983 *Metropolis and province: science in British culture, 1780–1850* Hutchinson, London 231–56
- Naylor S** submitted Collecting quoits: fieldwork cultures in the history of Cornish antiquarianism *Cultural Geographies*
- Noall C** 1963 A history of Cornish mail- and stage-coaches D. Bradford Barton, Truro
- Ophir A and Shapin S** 1991 The place of knowledge: the spatial setting and its relation to the production of knowledge *Science in Context* special issue 4
- Pyenson L and Sheets-Pyenson S** 1998 *Servants of nature* W.W. Norton, New York
- Ralfs J** 1885 On the history of botany in Cornwall *Transactions of the PNHAS*, Vol. II. N.S. 1884–1888 William Brendon and Son, Plymouth 15–20
- Roderick G W and Stephens M D** 1972 *Scientific and technical education in nineteenth century England* David and Charles, Newton Abbot
- Ross C C** 1880a Minutes of council committee meeting held on 3 May *PNHAS Minute Books 1880–1889* PNHAS Archive, Morrab Library, Penzance no pages.
- Ross C C** 1880b Minutes of public meeting held on 8 May *PNHAS Minute Books 1880–1889* PNHAS Archive, Morrab Library, Penzance no pages
- Ross C C** 1880c Minutes of committee meeting held on 12 June *PNHAS Minute Books 1880–1889* PNHAS Archive, Morrab Library, Penzance no pages
- Rusnock A** 1999 Correspondence networks and the Royal Society, 1700–1750 *British Journal of the History of Science* 32 155–69
- Secord A** 1994a Science in the pub: artisan botanists in early nineteenth-century Lancashire *History of Science* 32 269–315
- Secord A** 1994b Corresponding interests: artisans and gentlemen in nineteenth century natural history *British Journal of the History of Science* 27 383–408
- Shapin S** 1972 The Pottery Philosophical Society, 1819–1835: an examination of the cultural uses of provincial science *Science Studies* 2 311–36
- Shapin S** 1988 The house of experiment in seventeenth century England *Isis* 79 373–404
- Shapin S** 1994 *A social history of truth: civility and science in seventeenth century England* Chicago University Press, Chicago
- Shapin S** 1998 Placing the view from nowhere: historical and sociological problems in the location of science *Transactions of the Institute of British Geographers* 23 5–12
- Shteir A B** 1996 *Cultivating women, cultivating science: Flora's daughters and botany in England 1760–1860* Johns Hopkins University Press, Baltimore
- Smith C and Agar J** 1998 eds *Making space for science: territorial themes in the shaping of knowledge* Macmillan, Basingstoke
- Soojung-Kim Pang A** 1996 Gender, culture, and astrophysical fieldwork: Elizabeth Campbell and the Lick Observatory–Crocker Eclipse expeditions in **Kuklick H and Kohler R E** eds *Science in the field* *Osiris* special issue 11 17–43
- Stafford B M** 1995 Desperately seeking connections: four scenes from eighteenth-century laboratory life *Ecumene* 2 378–98
- Stafford B M** 1996 *Good looking: essays on the virtue of images* MIT Press, Cambridge MA
- Swann M** 2001 *Curiosities and texts: the culture of collecting in early modern England* Pennsylvania University Press, Philadelphia
- Symons J, Bailly W E and Cornish T** 1888 Curators' report *Transactions of the PNHAS*, Vol. III. N.S. 1888–1892 William Brendon and Son, Plymouth 7–10

- Thrift N, Driver F and Livingstone D** 1995 The geography of truth *Environment and Planning D: Society and Space* 13
- Tregelles G F** 1887 Report *Transactions of the PNHAS, Vol. II. N.S. 1884–1888* William Brendon and Son, Plymouth 307–9
- Tregelles G F** 1890 Report *Transactions of the PNHAS, Vol. III. N.S. 1888–1892* William Brendon and Son, Plymouth 197–200
- Turk F A** 1959 Natural history studies in Cornwall (1700–1900) *Journal of the Royal Institution of Cornwall* 229–79
- Willan L R** 1880 Minutes of committee meeting held on 19 June *PNHAS Minute Books 1880–1889* PNHAS Archive, Morrab Library, Penzance no pages
- Withers C** 1998 Towards a history of geography in the public sphere *History of Science* 36 45–78
- Withers C** 2001 *Geography, science and national identity: Scotland since 1520* Cambridge University Press, Cambridge
- Yanni C** 1999 *Nature's museums: Victorian science and the architecture of display* Athlone Press, London